Journal of The Physical Society of Japan
Vol. 60, No. 4, April, 1991, pp. 1300–1305

Meissner Effect in $La_{2-x}Ba_xCuO_y$ as Functions of x and y

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(Received October 4, 1990)

The ac susceptibility was measured for the powdered samples of $La_{2-x}Ba_xCuO_y$ with various Ba and oxygen contents. Double maxima at $x=0.08$ and 0.16 and a large dip at $x=0.12$ were observed in both the Meissner volume fraction f and the superconducting transition temperature T_c' where the Meissner signals start to show a large drop. We have newly found that for $x=0.12$ the decrease of the oxygen concentration y leads to a rise of T_c' and an increase of f, suggesting that the decrease of y suppresses the low-temperature phase transition between orthorhombic and tetragonal structures which makes the superconductivity unstable around $x=0.12$. Furthermore, we observed that the onset temperature of the Meissner effect T_c^{os} of about 30 K is almost independent of x and y, while the superconducting transition becomes broad and f decreases with lowering T_c. These results indicate that some optimum composition which shows the intrinsic superconductivity exists in the range of $0.08 \leqq x \leqq 0.16$.

[high T_c superconductivity, LBCO, Ba composition dependence, oxygen contents, ac susceptibility, Meissner effect, orthorhombic-tetragonal transition]

§1. Introduction

Since the discovery of the superconductivity of the $La_{2-x}Ba_xCuO_y$ system with the critical temperature, T_c, of about 30 K by Bednorz and Müller,[1] a lot of studies[2–5] have been reported on the superconductivity in this system, especially on x (Ba concentration) dependence of T_c. They commonly showed the appearance of superconductivity for $0.06 \leqq x \leqq 0.22$ with the double maxima of T_c around $x=0.08$ and 0.16 and with the minimum of T_c around $x=0.12$. Furthermore, it was reported that in the region around $x=0.12$ this system undergoes the phase transition from orthorhombic to tetragonal at about 60 K on cooling.[3–5] This transition may destroy the superconductivity for $x \sim 0.12$. However, as far as the authors recognize, there have been no works investigating the oxygen content y dependence of the superconducting properties, in particular, the Meissner volume fraction f by focusing on how to deal with the effect of weak coupling between crystalline grains. Recently, inhomogeneous superconducting properties have been pointed out for the isostructural $La_{2-x}Sr_xCuO_y$ from a view point of concentration dependence of T_c and f.[6,7] Close observation of x and y dependences of T_c and f is an indispensable key to clarify the behavior of the superconducting properties in $La_{2-x}Ba_xCuO_y$ as well as in $La_{2-x}Sr_xCuO_y$.

In this paper, we report the ac susceptibility ($ac-\chi_g$) measurements for the powdered samples of $La_{2-x}Ba_xCuO_y$ with various x and y, together with the composition dependence of T_c and f. As a result, it is suggested that as in the case of $La_{2-x}Sr_xCuO_y$ system, the sample showing the intrinsic superconductivity exists at an optimum concentration in the range of $0.08 \leqq x \leqq 0.16$, if the low-temperature structural phase transition is completely suppressed. Furthermore, we found that the decrease of y recovered the superconductivity for the sample with $x=0.12$, although we obtained the opposite results for the other contents.

§2. Experimentals

The samples of $La_{2-x}Ba_xCuO_y$ were prepared by solid state reactions of La_2O_3, $BaCO_3$ and CuO with 99.99% purities. Powders were ground, pelletized and calcined in air 900°C during two days for both decarbonisation and provisional reaction. After

that the same process was repeated several times at 1050~1100°C for two days. The obtained pellets were ground into fine powders with radii of 1~20 μm and then divided into three parts, each of which was heated in flowing O_2 or N_2 gas with at least 99.999% purity for 20 hours by the following condition: O_2 at 800°C (1), N_2 at 900°C (2) or N_2 at 950°C (3). Finally, each powdered sample was cooled slowly down to room temperature at a rate of 1~2°C/min in the same atmosphere. Hereafter we refer these samples as the sample (1), (2) and (3). All the samples were confirmed to be in a single phase with K_2NiF_4 type structure. The oxygen content y was measured by using the carbon-reduction method (Horiba, EMGA-550) and thermogravimetric analysis.

The real part of the complex susceptibility ac-χ_g was measured by use of conventional Hartshorn bridge method under the magnetic field of 10 mOe oscillating at frequency of 270 Hz. The measurements were performed from liquid He temperature to room temperature at a heating rate of 0.1~0.2 K/min. The samples for the measurements of ac-χ_g were soaked in paraffin in order to insulate powder particles each other. By this treatment, we can exclude the shielding effect due to the weak coupling among powder particles by the Josephson effect.

§3. Results and Discussion

The lattice parameters a, b and c determined by X-ray diffraction are shown in Fig. 1. Here, the symbols a^* and b^* are related to the lattice parameters a and b of the orthorhombic phase as $a^*=a/\sqrt{2}$ and $b^*=b/\sqrt{2}$. The structural transition between orthorhombic and tetragonal was confirmed for $x=0.06$ at room temperature: orthorhombic for $x<0.06$ and tetragonal for $x>0.06$. This structural transition, which can be seen also in $La_{2-x}Sr_xCuO_y$, is different from the similar transition at lower temperatures observed in this system as described below. The relation between x and y for various heat treatments is shown in Fig. 2. Roughly speaking, the values of y for the samples (1), (2) and (3) are about 4 and do not show large changes up to $x=0.1$. Beyond $x=0.15$, y of the sample (1) decreases almost

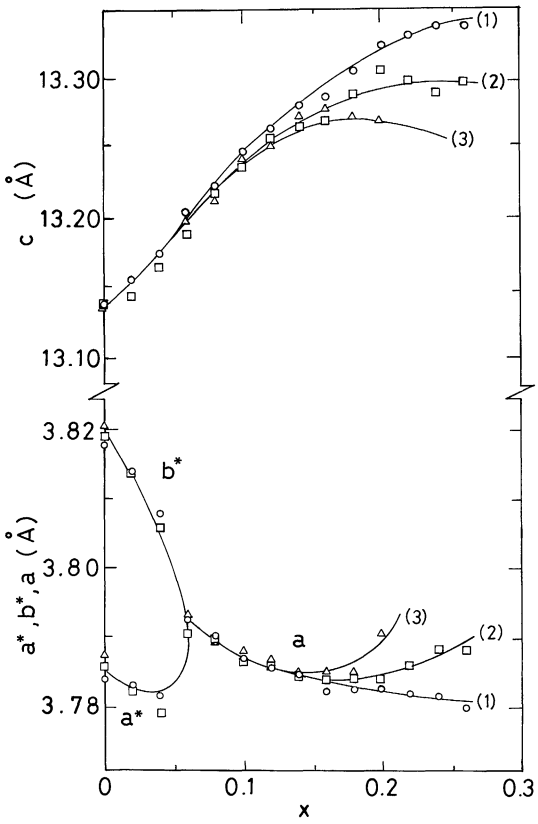


Fig. 1. Ba concentration dependence of lattice parameters, a, a^*, b^* and c in $La_{2-x}Ba_xCuO_y$ prepared by the heat treatments in pure gas flowing shown by the symbols as follows, ○: O_2 at 800°C (1), □: N_2 at 900°C (2) and △: N_2 at 950°C (3). The symbols a^* and b^* are related to the lattice parameters a and b of the orthorhombic phase as $a^*=a/\sqrt{2}$ and $b^*=b/\sqrt{2}$.

linearly and in parallel with the condition of ionic charge neutrality, $y=4-x/2$, as shown by the solid linear line. Here, the difference between the experimental values and this solid line corresponds to the half value of the doped hole content. The values of y for the samples (2) and (3), which were exposed to the reduction atmosphere, show considerable decreases but are between the case of the sample (1) and $y=4-x/2$.

In Figs. 3(a) and 3(b) we show the typical results of ac-χ_g measurements on the sample (1). One can find that the samples with $0.06\leqq x\leqq 0.22$ indicate the Meissner effect, leading to the occurrence of the bulk superconductivity in this concentration region. Simi-

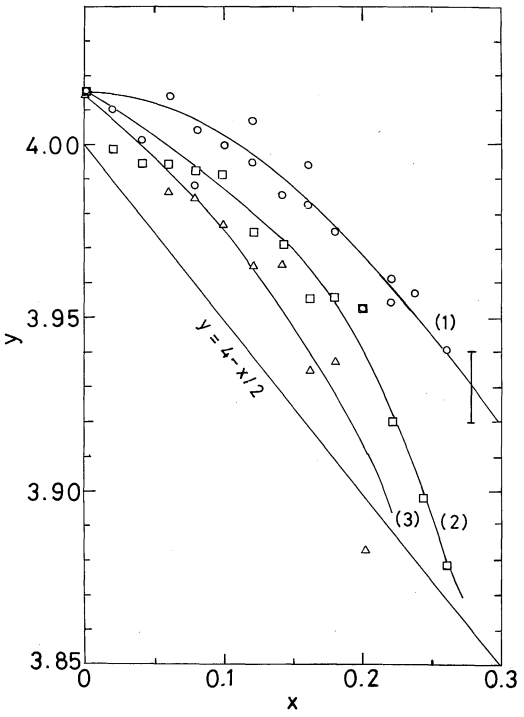


Fig. 2. Oxygen content y plotted against Ba concentration x in $La_{2-x}Ba_xCuO_y$ exposed to the various heat treatments. The symbols represent as follows, ○: (1), □: (2) and △: (3). The straight line shows $y=4-x/2$. Other curves are guides for the eye.

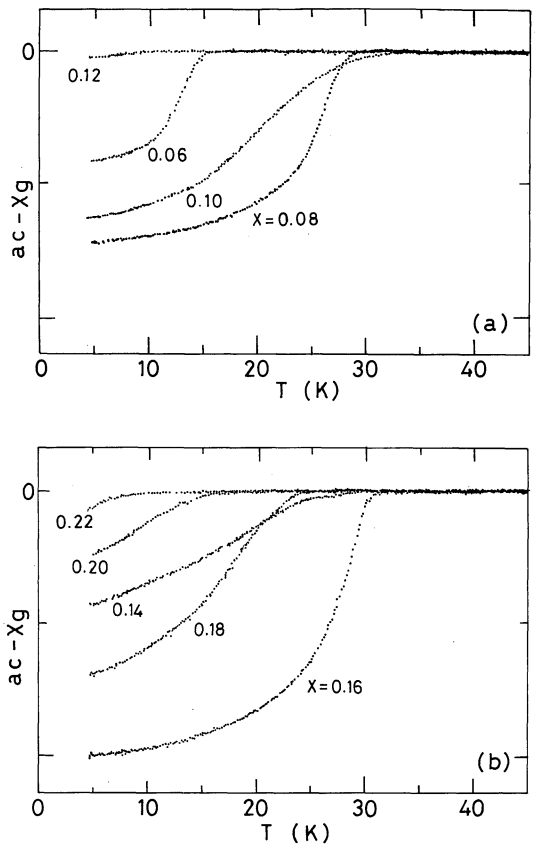


Fig. 3. (a), (b) Temperature dependence of $ac-\chi_g$ of $La_{2-x}Ba_xCuO_y$ prepared in the oxidation atmosphere (the sample (1), see text). Since the unit of the vertical axis is proportional to the induction voltage normalized by the sample weight, the values of $ac-\chi_g$ of different samples can be comparable with each other.

larly to the results of T_c previously reported,[2,4] a large dip has been observed at $x=0.12$ in the critical temperature T_c' determined as the temperature where the $ac-\chi_g$ starts to show a large drop (see the broken lines in Fig. 4 described below). From these figures, we could deduce the following two findings. One is that the sharpness of the transition is closely correlated with T_c': the lower temperature the value of T_c' shifts to, the broader the superconducting transition becomes. The other is that the Meissner effect starts to take place at T_c^{os} (the suffix os implies "on set"), being much higher temperature than T_c' and that T_c' approaches T_c^{os} with the increase of the Meissner volume fraction f (see also Figs. 6 and 7 presented below). As typical demonstrations, we show the enlarged plots near T_c for $x=0.08$, 0.10, 0.12, 0.14 and 0.16 in Fig. 4. It can be seen that T_c^{os} is almost independent of x. These results are quite similar to those obtained in the $La_{2-x}Sr_xCuO_y$ system except for the large dip of T_c' at $x=0.12$. In order to investigate the dependence of superconductivity on y, we measured $ac-\chi_g$ of the samples (2) and (3). As shown in Figs. 5(a) and 5(b) for the samples with $x=0.08$ and 0.18, T_c' decreases as (1)→(2)→(3) in order similarly to the case of $La_{2-x}Sr_xCuO_y$.[6] We obtained the similar results for most of the samples with other compositions. For $x\sim0.12$, however, T_c' increases as (1)→(2)→(3) in order, whereas T_c^{os} does not show any large change as shown in Fig. 5(c). All the data of T_c' and T_c^{os} are summarized in Fig. 6.

We determined the relative values of the Meissner volume fraction f from the values of $ac-\chi_g$ at 4.2 K. Although the absolute values of f can not be obtained directly from $ac-\chi_g$,

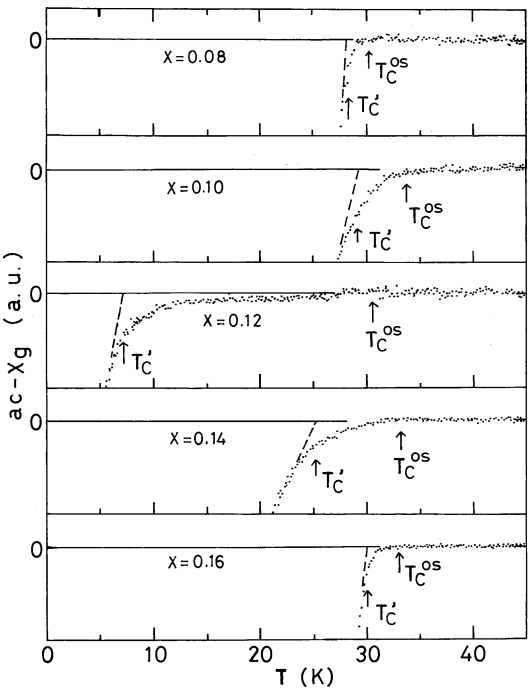


Fig. 4. Temperature dependence of $ac-\chi_g$ of the samples (1) with $x=0.08$, 0.10, 0.12, 0.14 and 0.16 near T_c. Here, T_c' and T_c^{os} defined in the text are represented by the arrows.

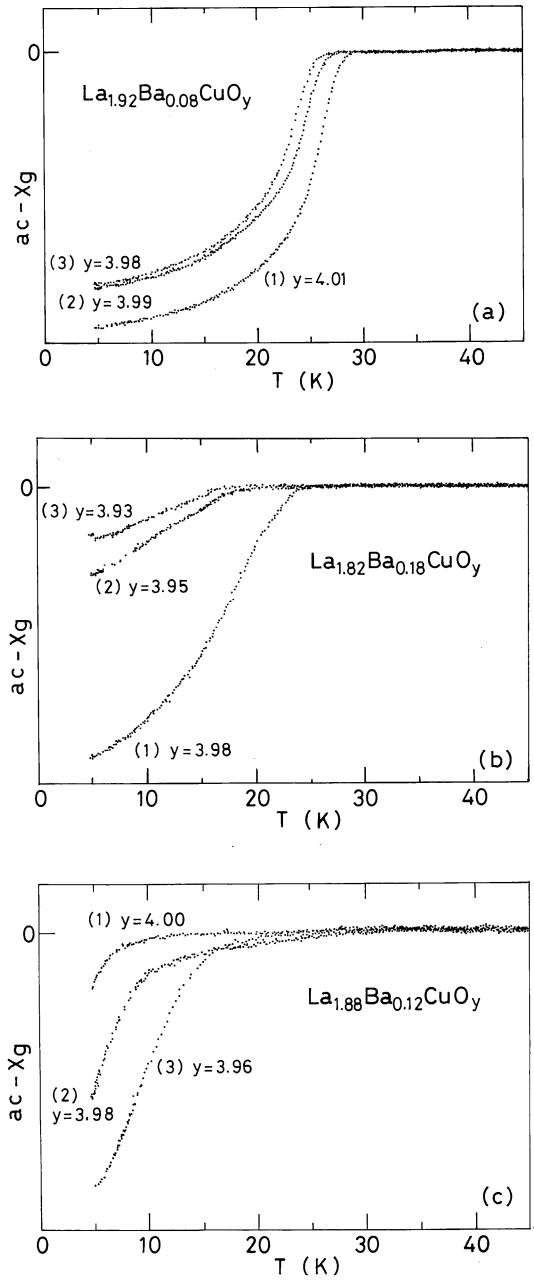


Fig. 5. (a), (b), (c) Temperature dependence of $ac-\chi_g$ of the samples (1), (2) and (3) with $x=0.08$(a), 0.18(b) and 0.12(c).

we confirmed that the sample (1) with $x=0.16$ showed the largest minus value of $ac-\chi_g$ corresponding to at least 70% of the ideal Meissner susceptibility, which was estimated in comparing with $ac-\chi_g$ of a powdered sample of lead metal with 6N purity. The concentration dependence of f is summarized in Fig. 7. In the sample (1) which was exposed to the oxidation atmosphere, the value of f has a local dip at $x=0.12$ between two maxima at $x=0.08$ and 0.16, which is also observed in the T_c' vs. x curves in Fig. 6. It was reported that around $x=0.12$ the structural transition from orthorhombic to tetragonal on cooling exists at low temperatures.[3-5] This phase transition may suppress the superconductivity in this x-region. Recently, we observed the anomalies of specific heats for the sample (1) with $x=0.12$ at about 220 K and 60 K.[8] These anomalies are attributed to the successive structural transformations (tetragonal→orthorhombic→tetragonal with decreasing temperature). When the oxygen is removed by the reduction treatment, i.e., in the cases of (2) and (3), the values of f decrease substantially except for $x\sim0.12$. The oxygen deficiency generally causes the instability of the superconductivity in the hole-doped systems as La 2-1-4

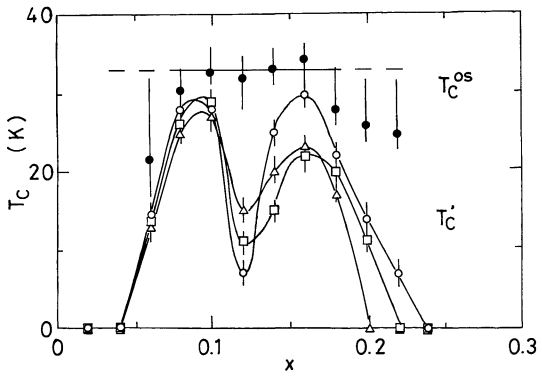


Fig. 6. Ba concentration x dependence of the superconducting transition temperatures T_c' and T_c^{os}. The symbols represent as follows, ○: (1), □: (2), △: (3) for T_c' and ●: (1), (2) and (3) for T_c^{os}.

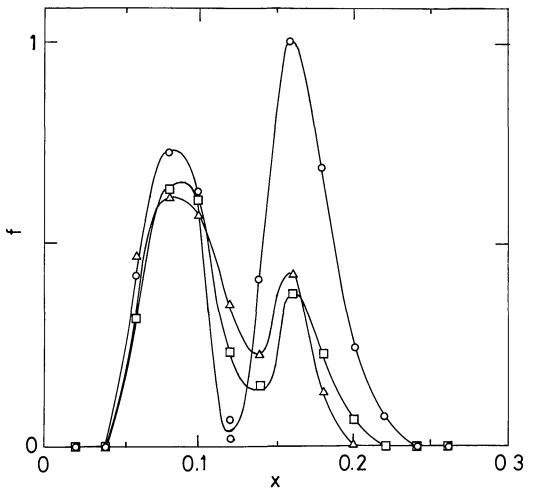


Fig. 7. Ba concentration x dependence of the normalized Meissner volume fraction $f=[ac-\chi_g(x)]/(ac-\chi_g(0.16)]$ at 4.2 K. The symbols represent as follows, ○: (1), □: (2) and △: (3).

and YBCO systems. Nevertheless, f as well as T_c' increases around $x\sim0.12$ with decreasing y when the sample is exposed to the heat treatment in the reduction atmosphere. Therefore, the oxygen deficiency restores the superconductivity for $x\sim0.12$. These results suggest that the decrease of y suppresses this low-temperature structural transformation around $x=0.12$.

Consequently, we found from above results that there exists the apparent correlation between T_c' and f: with lowering T_c', the superconducting transition becomes broad and f decreases. Furthermore, T_c^{os} does not depend upon x and y. These results indicate that the sample showing the intrinsic superconductivity exists at a certain optimum composition in the range of $0.08\leqq x\leqq 0.16$. For the other contents the Meissner signals mainly come from the weak-link couplings among superconductive crystalline grains involved inside a single powder particle, although the weak couplings among powder particles are almost excluded by insulating the particles each other using paraffin. Hence, the superconductivity takes place at $T\sim33$ K in a small amount of the intrinsic superconductive phase which is involved by fluctuations of concentrations and/or microscopic phase separation. The bulk superconductivity does not occur until the superconductive parts are joined with each other by weak couplings. Therefore, the drop of T_c' may be caused by the proximity effect.

Unfortunately, we have not yet identified the composition showing the intrinsic superconductivity, because the superconductivity is collapsed with the occurrence of the low-temperature structural transition around $x=0.12$. Recently, we have investigated the normal state behavior of ^{63}Cu nuclear spin-lattice relaxation rate $1/T_1$ for the sample (1) with $0.08\leqq x\leqq0.16$, and have found that $1/T_1$ is very similar to that in the normal state of $La_{2-x}Sr_xCuO_y$ with $0.08\leqq x\leqq0.16$ and $y\sim4.0$.[9] It has been found that there exists a relation between $1/T_1$ and T as $1/T_1=a+bT$, where a and b are constants.[9-11] The first term a is ascribed to ^{63}Cu spin fluctuations: the constant a is related to the inverse of the correlation frequency ω_c of antiferromagnetic Cu d-spin fluctuations. Thus, the value of ω_c can be roughly estimated from a. It is noted that there exists a correlation between T_c and ω_c universally independent of materials.[9-11] Therefore, these results of $1/T_1$ for the present system indicate nearly the same electronic state in $La_{2-x}Ba_xCuO_y$ as in $La_{2-x}Sr_xCuO_y$, suggesting that $La_{2-x}Ba_xCuO_y$ possibly shows nearly the same T_c as that in $La_{2-x}Sr_xCuO_y$ in the composition range of $0.08\leqq x\leqq0.16$, unless the low-temperature structural transition takes place.

§4. Concluding Remarks

We observed a local dip around $x=0.12$ and the double maxima around $x=0.08$ and 0.16 in T_c' as a function of x in agreement with the composition dependence of T_c previously reported.[2,4] In the region around $x\sim0.12$ this system undergoes the phase transition from orthorhombic to tetragonal at about 60 K as reported previously.[3-5] This structural transformation may destroy the superconductivity in this system. We found that the superconductivity was restored by the reduction of y for $x\sim0.12$. Therefore, the reduction of y may suppress the low-temperature phase transition. The following problems still remain: how does the oxygen content y act on this structural transition and how does this structural transition affect the superconductivity in this system? In order to clarify them, the X-ray diffraction experiments at low temperatures on the samples with various y are needed and now in progress.

From the results of ac$-\chi_g$ measurements with considering $1/T_1$ measurements, the sample with a certain optimum composition in the range of $0.08\leqq x\leqq0.16$ possibly shows the intrinsic superconductivity, if the low-temperature orthorhombic-tetragonal transition is completely suppressed. The superconducting properties are very inhomogeneous for the other concentrations as in the case of $La_{2-x}Sr_xCuO_y$.[6] Hence, it may not be intrinsic to discuss the concentration dependence of T_c in this system as well as $La_{2-x}Sr_xCuO_y$[6] in terms of the doped hole concentration, contrary to the previous reports.[12-16] The present result suggests the existence of fluctuations of contents and/or microscopic phase separation. Microscopic investigations, for example NQR, NMR and TEM, are now in progress.

Acknowledgment

The authors thank Professor H. Yasuoka, Dr. T. Imai, Dr. S. Kanbe and Professor Y. Ueda for helpful discussions. They are grateful to Dr. K. Ban (Horiba Ltd.) for oxygen analysis. This work is supported by a Grant-in-Aid for Scientific Research on Priority Areas "Chemistry of New Superconductors" (01645004 and 02227102) from the Ministry of Education, Science and Culture.

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